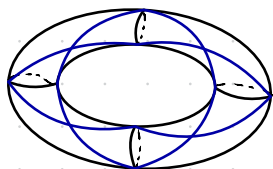
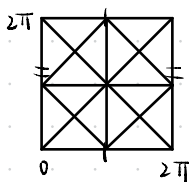


2. $\chi(\text{Klein} \# S^2) = \chi(\text{Klein}) + \chi(S^2) - 2 = 0 + 2 - 2 = 0$

3. $r(u, v) = (1 + \cos u) \cos v, (1 + \cos u) \sin v, \cos u) \quad u \in (0, 2\pi) \quad v \in (0, 2\pi)$



4. Girard 公式

$$S = (A + B + C) - \pi$$

$$\int_{\Delta} K dA = (A + B + C) - \pi \quad K=1$$

$$S = \int_{\Delta} dA = (A + B + C) - \pi$$

5. 反证法

$$\int K dA = 2\pi \chi(S)$$

$$g(S) = 10$$

$$\chi(S) = 2 - 2g(S) = -18$$

$$\therefore \int K dA = -36\pi$$

$$\text{假设: } \forall p \in S \quad K_p \geq -11\pi$$

$$\int K dA \geq -33\pi \neq -36\pi \quad \text{矛盾}$$